

SERVER SIDE MODULE

CONTENTS

This module has the following files:

1. SERVER MODULE PHASE 1.docx
2. SERVER MODULE PHASE 1.pdf
3. SERVER MODULE PHASE 1 MEDIA.zip

INTRODUCTION

The client has approached you to build a backend system and frontend web application for a **Financial Management Application** called **PintarMenabung**. The system must allow users to register, manage their wallets, record income and expenses, and view basic financial reports.

PROJECT OVERVIEW

This project consists of two main phases: In Phase 1, you are required to develop a **REST API** using the **Laravel Framework**, implementing core features such as user authentication, wallet management, transaction recording and financial reporting. In Phase 2, you will build a **frontend web application** that integrates with the API, using either **React** or **Vue** as your framework of choice.

To support your work, media files have been provided, including Postman Collection and Environment, a Database Dump (.sql), and Framework Scaffolding.

PHASE 1 - Backend Development (REST API)

A. Authentication

There are three endpoints: register, login, and logout. **Sanctum** should be used to create tokens. The generated token will be used to access all available endpoints. The logout function should only deactivate the token from the specific device making the request.

A1: Register

POST /api/auth/register

For users to create an account in the application.

Request Headers

Accept	application/json
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Body (JSON)

full_name	required.
email	required, valid email format, unique.
password	required, min 6 chars.

A1a: Response Success

Status Code	201
Body	<pre>{ "status": "success", "message": "Registration successful", "data": { "name": "Dedi", "email": "dedi@webtech.id", "updated_at": "2025-07-11T16:01:46.000000Z", "created_at": "2025-07-11T16:01:46.000000Z", "id": 6, "token": "1 wwu0rgoVe4Xyj5dvUmnf6WTPTaA..." } }</pre>

A1b: Response Invalid Field

Status Code	422
Body	<pre>{ "status": "error", "message": "Invalid field", "errors": { "name": ["The name field is required."], "email": ["The email has already been taken."], "password": ["The password field must be at least 6 characters."] } }</pre>

A2: Login

POST /api/auth/login

For users login into the application using email and password.

Request Headers

Accept	application/json
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Body (JSON)

email	required.
password	required.

A2a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Login successful", "data": { "id": 1, "name": "Budi", "email": "budi@webtech.id", "created_at": "2025-07-11T16:01:38.000000Z", "updated_at": "2025-07-11T16:01:38.000000Z", "token": "2 0Q2TAdua0F0s\lp1F4cxfrzQgkp ..." } }</pre>

A2b: Response Failed

Status Code	401
Body	<pre>{ "status": "error", "message": "Username or password incorrect" }</pre>

A3: Logout

POST /api/auth/logout

For logged-in user logout from the application. It specifically deactivates the authentication token associated with the current device making the request.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

A3a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Logout successful" }</pre>

A3b: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

B. Currency and Category

The application uses a Category system to classify financial transactions. Each category must be assigned a type, which can either be:

- Income – for transactions that increase the wallet balance (e.g., salary, bonus)
- Expense – for transactions that decrease the wallet balance (e.g., groceries, transportation)

Categories help users organize their finances and are required when creating a transaction.

B1: Get All Currencies

GET /api/currencies

For logged-in users to get all available currencies.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

B1a: Response Success

Status Code	200
Body	<pre>{ "message": "Get all currencies successful", "data": { "currencies": [{ "id": 1, "name": "US Dollar", "symbol": "\$", "code": "USD", "created_at": "2025-07-11T16:01:37...", "updated_at": "2025-07-11T16:01:37..." }, ...] } }</pre>

B1b: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

B2: Get All Categories

GET /api/categories

For logged-in users to get all available categories.

Request Headers

Accept	application/json
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Authorization	Bearer <TOKEN>
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B2a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Get all categories successful", "data": { "categories": [{ "id": 1, "name": "Outgoing Transfer", "icon": "↑", "type": "EXPENSE", "created_at": "2025-07-11T16:01:38 ...", "updated_at": "2025-07-11T16:01:38 ..." }, ...] } }</pre>

B2b: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

C. Wallet

A Wallet is a financial container used to track a user's income and expenses. Users can create multiple wallets to organize their finances (e.g., personal, savings, or business wallets).

Each wallet must display a current balance, which is automatically and accurately calculated based on all associated transactions.

- Income transactions will increase the balance.
- Expense transactions will decrease the balance.

The system must ensure that the wallet balance is always up-to-date and reflects the correct total from all transactions made within that wallet.

C1: Add Wallet

POST /api/wallets

For logged-in users to add a wallet.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

Body (JSON)

name	required.
currency_code	required, valid currency_code data.

C1a: Response Success

Status Code	201
Body	<pre>{ "status": "success", "message": "Wallet added successful", "data": { "name": "Savings", "user_id": 1, "updated_at": "2025-07-11T16:22:44.000000Z", "created_at": "2025-07-11T16:22:44.000000Z", "id": 26, "currency_code": "IDR" } }</pre>

C1b: Response Invalid Field

Status Code	422
Body	<pre>{ "status": "error", "message": "Invalid field", "errors": { "name": ["The name field is required."], "currency_code": ["The selected currency code is invalid."] } }</pre>

C1c: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

C2: Update Wallet

PUT /api/wallets/:walletId

For logged-in users to update their own wallet.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

Body (JSON)

name	required.
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C2a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Wallet updated successful", "data": { "id": 1, "user_id": 1, "name": "Cash IDR", "created_at": "2025-07-11T16:01:38.000000Z", "updated_at": "2025-07-11T16:26:25.000000Z", "deleted_at": null, "currency_code": "IDR" } }</pre>

C2b: Response Not Found

Status Code	404
Body	<pre>{ "status": "error", "message": "Not found" }</pre>

C2c: Response Forbidden

Status Code	403
Body	<pre>{ "status": "error", "message": "Forbidden access" }</pre>

C2d: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

C3: Delete Wallet

DELETE /api/wallets/:walletId

For logged-in users to delete their own wallet.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

C3a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Wallet deleted successful" }</pre>

C2b: Response Not Found

Status Code	404
Body	<pre>{ "status": "error", "message": "Not found" }</pre>

C2c: Response Forbidden

Status Code	403
Body	<pre>{ "status": "error", "message": "Forbidden access" }</pre>

C2d: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

C4: Get All Wallets

GET /api/wallets

For logged-in users to get all of their own wallets.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

C4a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Get all wallets successful", "data": { "wallets": [{ "id": 2, "user_id": 1, "name": "Bank", "created_at": "2025-07-11T16:01:38...", "updated_at": "2025-07-11T16:01:38...", "deleted_at": null, "currency_code": "IDR", "balance": 12250000 }, ...] } }</pre>

C4b: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

C5: Get Detail Wallet

GET /api/wallets/:walletId

For logged-in users to get details about their own wallets.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

C5a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Get detail wallet successful", "data": { "id": 1, "user_id": 1, "name": "Cash IDR", "created_at": "2025-07-11T16:01:38.000000Z", "updated_at": "2025-07-11T16:26:25.000000Z", "deleted_at": null, "currency_code": "IDR", "balance": 4865000 } }</pre>

C5b: Response Forbidden

Status Code	403
Body	<pre>{ "status": "error", "message": "Forbidden access" }</pre>

C5c: Response Not Found

Status Code	404
Body	<pre>{ "status": "error", "message": "Not found" }</pre>

C5c: Response Forbidden

Status Code	403
Body	<pre>{ "status": "error", "message": "Forbidden access" }</pre>

C5d: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

D. Transaction

A Transaction represents a financial record that occurs within a specific wallet. Each transaction must be linked to a wallet and assigned to a category, which determines whether it is an income or an expense.

Key attributes of a transaction include:

- Wallet – the wallet where the transaction belongs
- Category – defines the type (income or expense) based on the selected category
- Amount – the monetary value of the transaction (positive number)
- Date – the date when the transaction occurred
- Note (optional) – a short note about the transaction

The transaction's impact on the wallet balance depends on the type defined by the selected category:

- Income category → increases the wallet balance
- Expense category → decreases the wallet balance

The system must ensure that transactions are accurately recorded and that wallet balances are updated accordingly.

D1: Add Transaction

POST /api/transactions

For logged-in users to add a transaction.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

Body (JSON)

wallet_id	required, valid wallet_id data.
category_id	required, valid category_id data.
amount	required, integer, greater or equal 1
date	required, date format "Y-m-d"
note	optional

D1a: Response Success

Status Code	201
Body	<pre>{ "status": "success", "message": "Transaction added successful", "data": { "category_id": 3, "wallet_id": 1, "amount": 50000, "note": "starbucks", "date": "2025-07-31", "updated_at": "2025-07-11T16:44:09.000000Z", "created_at": "2025-07-11T16:44:09.000000Z", "id": 842 } }</pre>

D1b: Response Invalid Field

Status Code	422
Body	<pre>{ "status": "error", "message": "Invalid field", "errors": { "name": ["The name field is required."], "currency_code": ["The selected currency code is invalid."] } }</pre>

D1c: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

D2: Delete Transaction

DELETE /api/transactions/:transactionId

For logged-in users to delete their own transaction.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

D2a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Transaction deleted successful" }</pre>

D2b: Response Not Found

Status Code	404
Body	<pre>{ "status": "error", "message": "Not found" }</pre>

D2c: Response Forbidden

Status Code	403
Body	<pre>{ "status": "error", "message": "Forbidden access" }</pre>

D2d: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

D3: Get Transactions

GET /api/transactions

For logged-in users to get all of their own transactions ordered by transaction date descending.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

Request Parameters

page	optional, integer, default 1
per_page	optional, integer, default 25
month	optional, integer, between 1 and 12
year	optional, integer

D3a: Response Success

Status Code	200
Body	<pre>{ "current_page": 1, "data": [{ "id": 842, "category_id": 3, "wallet_id": 1, "amount": 50000, "note": "starbucks", "date": "2025-07-31", "created_at": "2025-07-11T16:44:09.000000Z", "updated_at": "2025-07-11T16:44:09.000000Z", "wallet": { "id": 1, "user_id": 1, "name": "Savings IDR",</pre>

	<pre> "created_at": "2025-07-11T16:01:38.000000Z", "updated_at": "2025-07-11T16:26:25.000000Z", "deleted_at": null, "currency_code": "IDR" }, "category": { "id": 3, "name": "Food & Drinks", "icon": "🍔", "type": "EXPENSE", "created_at": "2025-07-11T16:01:38.000000Z", "updated_at": "2025-07-11T16:01:38.000000Z" } }, ...], "from": 1, "last_page": 170, "per_page": 1, "to": 1, "total": 170 } </pre>
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D3b: Response Invalid Token

Status Code	401
Body	<pre> { "status": "error", "message": "Unauthenticated." } </pre>

E. Financial Report

E1: Get Summary by Expense Category

GET /api/reports/summary-by-category/expense

For logged-in users to get a summary by expense category.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

Request Parameters

month	optional, integer, between 1 and 12
year	optional, integer

E1a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Get summary by expense category successful", "data": { "summary": [{ "category": { "id": 13, "name": "Groceries", "icon": "🛒", "color": "#55EFC4", "type": "EXPENSE", "created_at": "2025-07-28T02:37:46.000000Z", "updated_at": "2025-07-28T02:37:46.000000Z" }, "amount": 50000 }] } }</pre>

	<pre> ... } } </pre>

E1b: Response Invalid Token

Status Code	401
Body	<pre> { "status": "error", "message": "Unauthenticated." } </pre>

E2: Get Summary by Income Category

GET /api/reports/summary-by-category/income

For logged-in users to get a summary by income category.

Request Headers

Accept	application/json
Authorization	Bearer <TOKEN>

Request Parameters

month	optional, integer, between 1 and 12
year	optional, integer

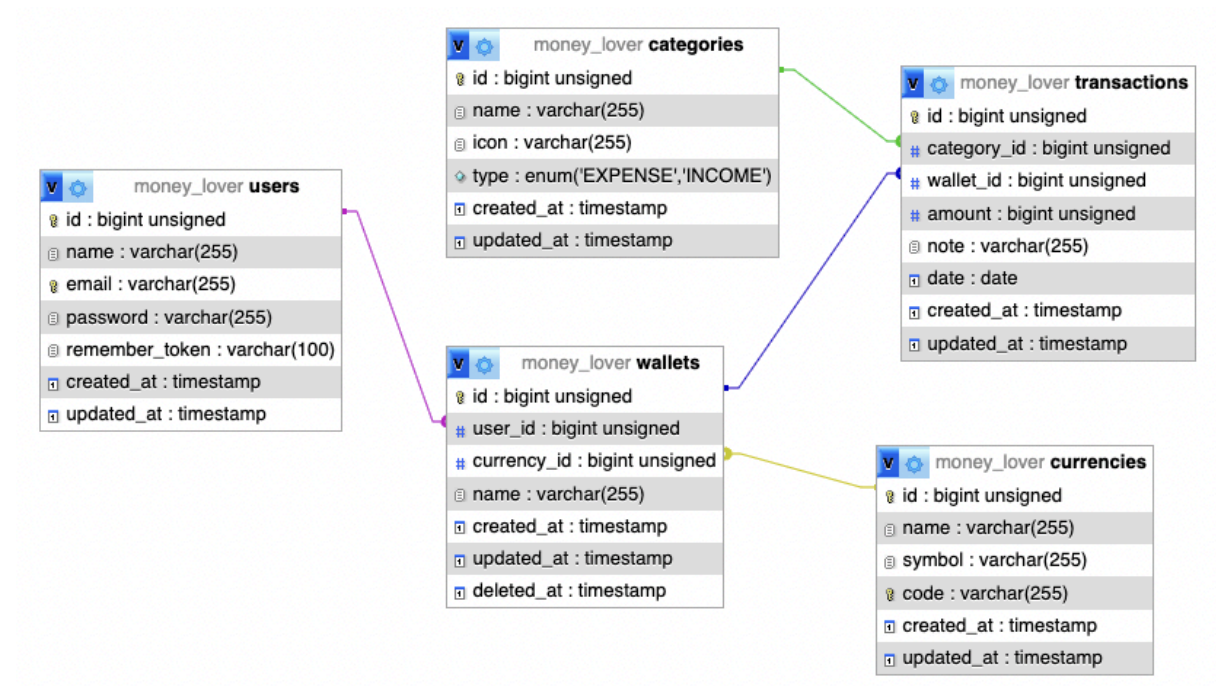
E2a: Response Success

Status Code	200
Body	<pre>{ "status": "success", "message": "Get summary by income category successful", "data": { "summary": [{ "category": { "id": 2, "name": "Incoming Transfer", "icon": "↓", "color": "#00B894", "type": "INCOME", "created_at": "2025-07-28T02:37:46.000000Z", "updated_at": "2025-07-28T02:37:46.000000Z" }, "amount": 1050000 }, ...] } }</pre>

E2b: Response Invalid Token

Status Code	401
Body	<pre>{ "status": "error", "message": "Unauthenticated." }</pre>

ER DIAGRAM



MEDIA FILES

- Postman Collection & Environment – contains all required API endpoints
- Database Dump (.sql) – includes initial schema and sample data
- Framework Installers – Laravel

INSTRUCTIONS

- You are not allowed to create additional endpoints.
- You can add extra fields to the response body if needed, **just make sure all required fields are included and the response structure is correct.**

PROJECT STRUCTURE & SUBMISSION

1. Create a root folder named **XX_SERVER_MODULE** where **XX** is your computer number.
2. Inside the root folder, organize your files as follows:
 - **XX_SERVER_MODULE/**
 - **BACKEND/** # Your backend project code
 - **db-dump.sql** # Exported database dump
 - **db-diagram.pdf** # ER Diagram
3. Zip the root folder into a single file:
XX_SERVER_MODULE_PHASE_1.zip
4. Exclude the following folders when zipping:
 - **vendor/**
 - **node_modules/**
5. Submit **XX_SERVER_MODULE_PHASE_1.zip** to the submission page.